INCH-POUND

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MILITARY SPECIFICATION

TANK AND PUMP UNIT, FUEL DISPENSING, FOR TRUCK MOUNTING AND SINGLE TANK UNIT, FUEL DISPENSING, 525 GALLON, TRAILER MOUNTED

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a fuel dispensing tank and pump unit for truck mounting and single tank unit, fuel dispensing for trailer mounting (see 6.1).

1.2 Classification. The units covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Tank and Pump Unit, Fuel Dispensing, for Truck Mounting.
Type II - Single Tank Unit, Fuel Dispensing, 525 Gallon, Trailer Mounted.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-B-636	-	Boxes , Shipping, Fiberboard.
PPP-T-60	-	Tape : Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of,
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-T-22238	-	Tank, Liquid Storage, Metal for Petroleum Products, Horizontal with Skids, 525 Gallon and 600-Gallon Capacity.
MIL-B-26195	-	Boxes , Wood Cleated, Skidded, Load-Bearing Base.
MIL-C-46168	-	Coating, Aliphatic Polyurethane Chemical Agent Resistant.
MIL-C-52950	-	Crates , Wood, Open and Closed.

STANDARDS

FEDERAL

FED-STD-595	-	Colors Used in Government Procurement.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
KIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirement for the Control of Electromagnetic Interference.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS3102	-	Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type.

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MS25043	- Cover, Electrical Connector, Receptacle, AN Type.
MS52131	- Conncetor, Plug , Electrical Intervehicle Power Cable.

2.1.2 Other Government drawings The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13226E2150	- Tank and Pump Unit, Fuel Dispensing, for Truck Mounting (EMD).
TA13228E4260	- Single Tank Unit, Fuel Dispensing, 525 Gallon, Trailer Mounted.
13217E2987	- Drum Unloader Suction Stub, Type II.
13217E7131	- Motor, Electric, 1.5 HP, 24 VDC.
13217E7135	Pump Assembly, Electric Motor Driven.
13225E8410	- Kit, Flow Meter, Liquid, Pump Unit.
13226E2148	- Pump Unit, Liquid Dispensing, for Truck Mounting.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center. AlTN: STRBE-FSH, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017).

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.2 - Structural Welding Code - Aluminum.

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(Application for copies should be addressed to the American Welding Society, 550 North LeJeunne, P.O. Box 351040, Miami, FL 33125.)

HYDRAULIC INSTITUTE

Hydraulic Institute Standards.

(Application for copies should be addressed to the Hydraulic Institute, 2130 Keith Building, Cleveland, OH 44115.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210,)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The tank and pump unit and the single tank unit, fuel dispensing for trailer mounting, (hereinafter called "dispensing unit5") shall be in accordance with the applicable top assembly drawings and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances

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is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the dispensing unit. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures. etc.) prepared by be contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to the following provisions.

3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Identification of materials and finishes, The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.3 Recovered materials. For the purpose of this requirement. recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the dispensing units may be newly fabricated from recovered materials to the maximum extent practicable, provided the dispensing units produced meets all other requirements of this specification. Used, rebuilt or remanufactured components , pieces and parts shall not be incorporated in the dispensing units .

3.4 Environmental conditions.

3.4.1 Operatining temperature . The dispensing units shall perform as specified in any ambient temperature from -25 to +140 “F.

3.4,2 Storage temperature. The dispensing units shall not be damaged by storage in any ambient temperature from -50 to +160 “F.

3.5 Transportability. The dispensing units shall be transportable under the following conditions without evidence of malfunction, permanent deformation, or damage to the tank, pump, or components:

a. Over hard-surfaced roads at an average speed of 50 miles per hour (mph) .

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b. Over gravel roads at an average speed of 20 mph
c Over cross-country terrain at an average speed of 15 mph.

3.5.1 Slope capability. The dispensing units shall be able to travel on slopes up to 20 percent and up-grade and down-grade on a ramp of 30 percent slope without damage or shifting of the unit, or inability of the pump to operate at rated capacity,

3.5.2 Emergency stop. The dispensing units shall withstand the effects of emergency stops at forward speeds up to convoy speeds and at reverse speeds up to 10 mph without permanent deformation, failure (see 6.6.2), or damage to the mounting or securing systems and without malfunction or damage to the dispensing units, or failure to operate at rated capacity.

3.6 Type I dispensing unit. The type I dispensing unit shall conform to TA13226E2150 and as specified herein.

3.6.1 Reliability. The specified mean-time-between-failure (MTBF) shall be 114 hours when the dispensing unit is operated at rated capacity and tested as specified in 4.5.2.7.

3.6.2 Maintainability.

3.6.2.1 Maintenance ratio. The dispensing unit shall have a maintenance ratio, (see 6.6.1), of not more than 0.08. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.6.3 Pump assembly. The pump assembly shall conform to drawing 13217E7135 and as specified herein. The pump shall be centrifugal, self-priming, close coupled, with a required net positive suction head (NPSH) of not more than 8 feet when operating at rated speed, flow and head.

3.6.3.1 Hydrostatic pressure. The pump assembly shall withstand a hydrostatic pressure of not less than 100 pounds per square inch (psi) without evidence of leakage, rupture, or distortion when tested as specified in 4.5.2.9.

3.6.3.2 Priming. The pump assembly shall prime in not more than 2 minutes when tested as specified in 4.5.2.10.

3.6.3.3 Efficiency. The pump assembly shall develop an efficiency of not less than 55 percent when tested as specified in 4.5.2.11.

3.6.3.4 Capacity. The pump assembly shall deliver liquid hydrocarbon fuels having a specific gravity of 0.75 at not less than 50 gallons per minute (gpm) at a total head of 60 feet when tested as specified in 4.5.2.11,

3.6.4 Flow meter assembly. When specified (see 6.2), flow meter kit assembly conforming to drawing 13225E8410 shall be furnished and installed.

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3.6,5 Drum unloader suction stub . When specified (see 6.2), a drum unloader suction stub, type II in accordance with drawing 13217E2987 shall be furnished.

3.6.6 Pump unit.

3.6.6.1 Hydrostatic pressure. When tested as specified in 4.5.2,12.1 through 4.5.2.12.4, the individual components of the pump unit, shown on drawing 13226E2148, shall withstand a hydrostatic pressure of not less than 50 psi without leakage through any component or connection.

3.6.6.2 Air pressure. When tested as specified in 4.5.2.12.5, the complete pump unit, as assembled prior to shipment, shall withstand an air pressure test of not less than 10 psi without leakage through any component or connection.

3.6.6.3 Continuity. When tested as specified in 4.5.2.13, the pump unit shall have a continuous electrical circuit between the nozzle bonding wire and the ground wire bonding clip.

3.6.7 Electric motor The electric motor shall conform to drawing 13217E7131 and as specified herein.

3.6.7.1 Power cable receptacle~ The power cable to the motor shall be equipped with a receptacle conforming to HS52131-1.

3.6.7.2 Control cable receptacle. The control cable to the motor shall be equipped with a receptacle conforming to MS3102R16S-4S with cover conforming to MS25043-16DA.

3.6.7.3 Explosion proof. The electric motor assembly complete with all receptacles and connectors shall be explosion proof in accordance with the National Electrical Code NFPA No. 70, class 1, division 1, group D.

3.6.7.4 Electromagnetic interference. The electromagnetic interference emission characteristics of the dispensing unit shall conform to MIL-STD-461, class Cl.

3.6.7.5 Splashproof The electric motor assembly complete with all receptacles and connectors shall be splashproof, in accordance with NEMA-MG-1.

3.7 Treatment and painting. The portions of the dispensing units normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704. type F or C, as applicable. Unless otherwise specified (see 6.2), topcoat color shall be green 383 (no. 34094 of FED-STD-595) conforming to MIL-C-46168. Mating surfaces and surfaces in contact with fuel shall not be painted.

3.8 Identification q arking. The dispensing units shall be identified in accordance with HIL-STD-130.

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3.9 Type II dispensing unit. The type II dispensing unit shall conform to TA13228E4260 and as specified herein.

3.10 Workmanship.

3.10.1 Casting and forgings. All parts, components , and assemblies of the dispensing unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.10.2 Metal fabrication. Metal used in the fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape. Plasma cutting may be used on aluminum alloys; however any heat affected zone generated by the cutting process must be removed prior to welding. Precaution shall be taken during all fabrication to avoid over-heating. External surfaces shall be free of burrs, sharp edges and comers except where sharp edges and comers are required or where they are not detrimental to safety.

3.10.3 Bolted connection. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.10.4 Machine work. Tolerances and gauges for metal fits shall conform to the limits specified on the applicable drawings.

3,10.5 Welders and welding operators. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by one of the welding codes for the types of welding operations to be performed and that such qualification is effective as defined by one of the codes which follow:

AWS D1.2, - ASME Boiler and Pressure Code - Section IX

Contractors who only make horizontal welds need not qualify welders for all position welding. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer’s representative.

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3.10.6 Welding Procedure . Welding procedures shall be qualified in accordance with AWS D1.2 or ASME Boiler and Pressure Vessel Code, Section IX and shall be made available to the contracting officer or designated representative . The appropriate qualified procedures shall be available at each welding station.

3.11 Human factors engineering. The dispensing units shall conform to the requirements of MIL-STD-1472. Special emphasis shall be given, but not limited to, MIL-STD-1472, paragraphs 4 (General Requirements), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.9 (Design for Maintainer), and 5.13 (Hazards and Safety), as applicable.

3.12 Safety. All shafts or rotating parts of the dispensing unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto electrical equipment. Adequate shielding may be used to attain the same purpose. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified In the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform arty of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements .

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined and tested in accordance with referenced specifications, standards, and drawings, as applicable.

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4.1.3 Disassembly inspection. Failure of any examination or test by the first article model shall be cause for disassembly, in the presence of a Government representative, of the first article model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-furnished property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article dispensing unit shall be examined as specified In 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article dispensing unit shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformante inspection.

4,4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2,1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified In 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the test marked “X” in column 2 of table I.

4.5 Inspection procedure.

4.5.1 Examination. The dispensing unit shall be examined as specified herein for the following defects:

101. Dimensions not as specified (see 3.1)

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102. Material not as specified herein or not as shown on applicable drawings (see 3.3).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3-1).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.2).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the dispensing unit (see 3.3.3).
106. When required, flowmeter missing or not as specified (see 3.6.4).
107. When required, drum unloader missing or not as specified (see 3.6.5).
108. Paint color not as specified herein or not as shown on drawings (see 3.7).
109. Treatment and painting not as specified herein or not as shown on drawings (see 3.7).
110. Identification markings missing, incomplete, or not legible as specified on applicable drawing (see 3.8).
111. Workmanship not as specified (see 3.10).
112. Welding procedure qualification or welder qualification not as specified (see 3.10.5 and 3.10.6).
113. Human factors engineering not as specified (see 3.11).
114. Safety not as specified (see 3.12).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable, provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92 inches of mercury and liquid fuel at 60 "F having a specific gravity of 0.75 and vapor pressure of 4.0 psi. Water at a temperature of between 60 to 80 "F may be used as the test fluid.

4.5.2.1.1 Test schedule.

TABLE I. Test schedule.

First article	Quality assurance sample	Tests	Test paragraph	Requirement paragraph
1	2	3	4	5
X		High temperature.	4.5.2.2	3.4.1 and 3.4.2
X		Low temperature.	4.5.2.3	3.4.1 and 3.4.2
X		Transportability.	4.5.2.4	3.5
X		Slope.	4.5.2.5	3.5.1

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TABLE I. Test schedule - continued.

First article	Quality assurance sample	Tests	Test paragraph	Requirement paragraph
1	2	3	4	5
X		Emergency stop.	4.5.2.6	3.5.2
X		Reliability.	4.5.2.7	3.6.1
X		Maintainability.	4.5.2.8	3.6.2
X	X	Hydrostatic.	4.5.2.9	3.6.3.1
X		Self-priming.	4.5.2.10	3.6.3.2
X	X	Pump capacity.	4.5.2.11	3.6.3.4
X		Pump efficiency.	4.5.2.11	3.6.3.3
X	X	Pump unit.	4.5.2.12	3.6.6.1 and 3.6.6.2
X	X	Continuity.	4.5.2.13	3.6.6.3
X		Explosion proof	4.5.2.14	3.6.7.3
X		Electromagnetic interference	4.5.2.15	3.6.7.4
X		Splashproof.	4.5.2.16	3.6.7.5

4.5.2.2 High temperature. The dispensing units shall be tested as specified in method 501.3 of MIL-STD-810. The maximum ambient storage temperature shall be 160 “F, and the maximum ambient operating temperature shall be 125 "F. The operating period of procedure I shall be not less than 2 hours. The dispensing units shall be operated at rated capacity. Nonconformance to 3.4.1 or 3.4.2 shall constitute failure of this test.

4.5.2.3 Low temperature. The dispensing units shall be tested as specified in method 502.3 of MIL-STD-810. The minimum ambient storage temperature shall be -50 “F. The lowest ambient operating temperature shall be -25 l F. The operating period of procedure I shall be not less than 2 hours. The dispensing units shall be operated at rated capacity. Nonconformance to 3.4.1 or 3.4.2 shall constitute failure of this test.

4.5.2.4 Transportability. The type I dispensing unit shall be driven and the type II dispensing unit shall be towed over a smooth, hard surfaced road at an average speed of 50 mph for 100 miles, and cross-country at an average speed of 15 q ph for a total of 200 miles, and over gravel roads at an average speed of 20 mph for a total of 100 miles. The tanks shall be full for 50

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percent, half full for 25 percent and empty for 25 percent of the mileage of each type surface. After each of the above segments of the test, the dispensing units shall be operated at rated capacity for 30 minutes. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.5 Slope. The dispensing units fully loaded shall be moved along a side slope of 20 percent. The procedure shall be repeated in the opposite direction. The truck shall be driven up-grade and down-grade on a ramp of 30 percent slope. After negotiating the ramps, the dispensing units shall be operated at rated capacity for not less than 1 hour. Nonconformance to 3.5.1 shall constitute failure of this test.

4,5.2.6 Emergency Stop. The type I dispensing unit shall be subjected to a series of emergency stops while fully loaded. The truck shall be stopped while moving forward at 10 q ph five consecutive emergency stops at each of the following speeds: 20, 30, and 45 mph forward and, finally 10 mph in reverse. The type I dispensing unit shall be operated at rated capacity for not less than 1 hour on completing the emergency stops. The fully loaded type II dispensing unit shall be subjected to a series of emergency stops following the same procedure as for type I dispensing unit, except that the emergency stops shall be at 10, 20, and 35 mph in the forward direction and at 5 mph in reverse for a total of 5 consecutive times at each speed. The mounting and tiedown systems shall be inspected throughout the both tests. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.7 Reliabia bility. The dispensing units selected for first article shall be tested as specified in 4.5.2.12 for 450 hours. Experiencing 6 or more failures during this test, shall constitute failure of this test. Operating time accumulated during testing as specified in 4.5.2.2, 4.5.2.3, 4.5.2.4, 4.5.2.5, and 4.5.2.6 shall be included in data for reliability calculations. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.8 Maintainability.

4.5.2.8.1 Maintenance ratio. The maintenance ratio shall be computed during production testing. Nonconformance to 3.6.2.1 shall constitute failure of this test.

4.5.2.9 Hydrostatic Pressure. Subject the pump assembly to a hydrostatic pressure of not less than 100 psi for a period of not less than 5 minutes. Nonconformance to 3.6.3.1 shall constitute failure of this test.

4.5.2.10 self-priming. Drain the suction hose and pump assembly and place the suction end of the hose in liquid to be pumped with the liquid level at not less than 8 feet below the impeller eye. Prime the pump with gasoline or cleaning solvent and start the pump. Nonconformance to 3.6.3.2 shall constitute failure of this test.

4.5.2.11 pump capacity. Test the pump assembly specified in 3.6.3.4 in accordance with the applicable sections of the centrifugal pump test code of the Hydraulic Institute. The test liquid may be water within the temperature

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range of 60 to 80 "F. Characteristic curves shall be plotted from this test while operating from no flow to maximum flow, plotting dynamic head, efficiency, power input, and rpm versus capacity in gpm The test data an characteristics curves shall include complete information for correlation of the test liquid to determine hydrocarbon fuel capacity specified in 3.6.3.4 and efficiency specified in 3.6.3.3. Nonconformance to 3.6.3.4 or 3.6.3.3 shall constitute failure of this test.

4.5.2.12 Pump unit.

4.5.2.12.1 Hose reel assembly. The hose reel with coupling halves shall be subjected to a hydrostatic pressure of not less than 50 psi for 20 minutes. Nonconformance to 3.6.6.1 shall constitute failure of this test.

4.5.2.12.2 Nozzle assembly. The nozzle with coupling half shall be subjected to a hydrostatic pressure of not less than 50 psi for 20 minutes. Nonconformance to 3.6.6.1 shall constitute failure of this test.

4.5.2.12.3 Filter separator assembly. The filter separator with coupling halves shall be subjected to a hydrostatic pressure of not less than 50 psi for 20 minutes. Nonconformance to 3.6.6.1 shall constitute failure of this test.

4.5.2.12.4 Hydrostatic pressure. The pump unit in accordance with drawing 13226E2148 shall be subjected to hydrostatic pressure of not less than 50 psi for 20 minutes. Nonconformance to 3.6.6.1 shall constitute failure of this test.

4.5.2.12.5 Air pressure. Subject the complete pump unit to an internal air pressure of 10 psig. With the air pressure source shut off, inability of the pump unit to maintain the 10 psig pressure for a period of five minutes shall constitute failure of this test. Nonconformance to 3.6.6.2 shall constitute failure of this test.

4.5.2.13 Continuity. The ground wire on the pump unit assembly shall be connected through an ohmmeter to each nozzle bonding wire to determine continuity. An ohmmeter reading greater than 10 ohms shall constitute failure of this test, Nonconformance to 3.6.6.3 shall constitute failure of this test.

4.5.2.14 Explosion proof. The electric motor assembly shall be UL listed for the service specified in 3.6.7.3.

4.5.2.15 Electromagnetic interference. The first article model shall be tested to determine conformance to 3.6.7.4. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Nonconformance to 3.6.7.4 shall constitute failure of this test.

4.5.2.16 Splashproof. The electric motor assembly shall be UL listed for the service specified in 3.6.7.5.

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4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.4. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Inspection stages. Inspection shall be in three stages, as follows:

a. The first stage shall include inspection of procedures, methods, and packaging marking.
b. The second stage shall include inspection of arrangement of components and the anchoring of the pump unit to the container base.
c. The third stage shall include inspection of closure, strapping, and marking of the completed pack as prepared for shipment.

4.6.2.2 Unit of product. For the purpose of inspection, a dispensing unit prepared for any applicable inspection stage shall be considered a unit of product.

4.6.2.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.4 Examination. Each sample pack shall be examined in accordance with 4.6.2.3 for the following major defects.

115. Unprotected surfaces not coated with preservative as specified for level A (see 5.2.1.1).
116. Tank and tank components not preserved as specified for level A or C (see 5.2.1.2 and 5.2.2.1).
117. Interior surfaces of the pump not coated with preservative for level A (see 5.2.1.3).
118. Opening into pump and motor not sealed with tape as specified for level A or C (see 5.2.1.3, 5.2.1.4, and 5.2.2.2).
119. Openings into filter-separator unit not sealed with tape as specified for level A or C (see 5.2.1.5 and 5.2.2.2).
120. Filter elements not preserved as specified for level A or C (see 5.2.1.6 and 5.2.2.2).
121, Consolidation not as specified for level A or C (see 5.2.1.10 and 5.2.2.2).
122. Packing not as specified for level A or C (see 5.3).
123. Marking illegible, incomplete, incorrect, or missing (see 5.4).

5. PACKAGING

5.1 First article pack The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove prior to

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starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4) . The first article pack shall be accomplished utilizing -the 'first article model dispensing unit. Any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government, to facilitate comparison between the first article dispensing unit and production dispensing units.

5.2 Preservation. The level of preservation shall be level A or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unprotected surfaces. Unpainted exterior metal surfaces of the dispensing units requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative conforming to the applicable specification listed in and applied in accordance with MIL-P-116.

5.2.1.2 Tanks. Tanks and tank components shall be preserved and consolidated in accordance with MIL-T-22238, level A.

5.2.1.3 Pump. Interior surfaces of the pump shall be coated with preservative lubricating oil conforming to type P-10, type I, grade 10, of MIL-P-116, applied by introducing the preservative into the pump intake while the pump is being actuated. Sufficient preservative shall be applied until the preservative is being discharged at the pump discharge opening. Excess preservative shall be allowed to drain. After the drainage ceases, all openings into the pump shall be sealed with tape conforming to PPP-T-60, type Iv.

5.2.1.4 Electric motor. Openings into the electric motor shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.5 Filter-separator The filter-separator shall not require preservation. Openings into the filter-separator unit shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.6 Filter el nts . Each filter element shall be inserted into a polyethylene bag fabricated from material conforming to L-P-378, type I, class 1, thickness .0040, and the bag closed by heat sealing. Each bagged element shall be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. The boxes shall be sealed in accordance with method V of the appendix to the box specification.

5.2.1.7 Technical Duplications . Technical publications for each dispensing unit shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3.

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5.2.1.8 Consolidation. The overpack kit items such as the preserved filter elements, extra fittings, repair parts, tools, and technical publications for each dispensing unit shall be consolidated together in the storage box provided on the unit. Cushioning shall be provided to prevent movement of the items within the storage box. The box lid shall be closed and secured.

5.2.2 Level C.

5.2.2.1 Tank units. Any openings into the tank shall be sealed with tape, PPP-T-60, type IV, to prevent entrance of water and dirt. Repair parts and tools shall be preserved in accordance with level A requirements of MIL-T-22238. Consolidation of accessories, parts and tools shall be in a weather-resistant box conforming to PPP-B-636, grade appropriate for the size and weight of the contents.

5.2.2.2 Pump units. All openings into the pump, electric motor, filter-separator shall be sealed with waterproof tape. Filter elements, repair parts, tools and technical publications shall be preserved as specified for level A. Consolidation shall be as specified for level A.

5.3 Packing. The level of packing shall be level A or C, as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Tank units. Each tank unit with tank components, preserved as specified in 5.2, shall be packed in accordance with MIL-T-22238 as specified for level A.

5.3.1.2 Pump Units. Each pump unit with pump unit components, preserved as specified in 5.2, shall be packed in a box conforming to MIL-B-26195, type II, style A, class 1. Blocking, bracing, anchoring and cushioning shall be in accordance with MXIL-STD-1186.

5.3.2 Level C.

5.3.2.1 Tank unit. The tank unit shall be shipped crated or uncrated. Consolidated items shall be placed and secured within the crate, if crated, or shipped concurrently with the tank. If possible, the consolidation box may be secured in a protected location on the tank assembly, if the tank is uncrated.

5.3.2.2 Pump unit. The pump unit shall be packed in a box as specified for level A or in an open crate conforming to MIL-C-52950, type V, style as appropriate for the weight of the contents. The consolidation box shall be secured within the shipping container.

5.4 Marking. In addition to any special marking specified in the marking contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The dispensing units are intended to convert general purpose cargo vehicles into bulk fuel transporters and refuelers for military vehicles and aircraft. The type I dispensing unit is intended for use with a 5 ton cargo trucks. The type II dispensing unit is intended for use with a 5 ton dual axle cargo trailer.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests shall be conducted by the contractor (see 3.2).
e. When flow meter assembly is required (see 3.6.4).
f. When drum unloader suction stub is required (see 3.6.5).
g. Color required if other then green (see 3.7).
h. Time frame required for submission of first article pack (see 5.1).
i. Level of preservation and packing required (see 5.2 and 5.3).
j. Any special marking (see 5.4).
k. Spare set of four (4) filter elements required (NSN 4330-00-983-0998).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for sample(s) for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 First artitle pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the dispensing units in accordance with this specification.

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6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.6 Definitions.

6.6.1 Maintenance ratio. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time.

6.6.2 Failure. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustments, repair, or replacement action using controls and on equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

6.7 Subject term (key word) listing.

Dispensing unit, liquid
Pump, centrifugal, 50 GPM, EMD
Tank, 525 gallon
Tank, 600 gallon
Tank and pump unit, truck mounting

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-ME

Preparing activity:
Army - ME

Project 4930-A363

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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-40136J(ME)	

3. DOCUMENT TITLE Tank and Pump Unit, Fuel Dispensing, For Truck Dispensing, 525 Gallon, Trailer Mounted

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewr

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORG
c. ADDRESS (Include Zip Code)	d. TEL (1) Co ((2) AU

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TEL (1) Com (703)
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU Defen 5203 Telep

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